

INTRODUCTION TO POWER BI

1. What is Power BI?

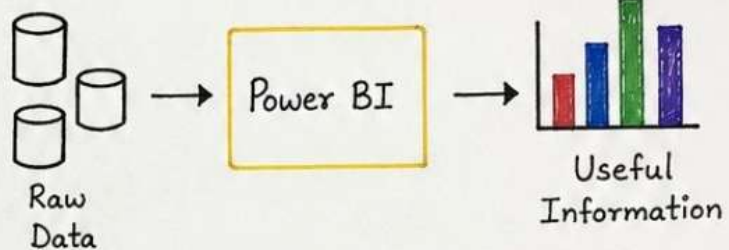
- Power BI is a Business Intelligence (BI) tool developed by Microsoft. It helps us to connect, transform, visualize and share data to make better business decisions.

2. Business Intelligence (BI) Basics

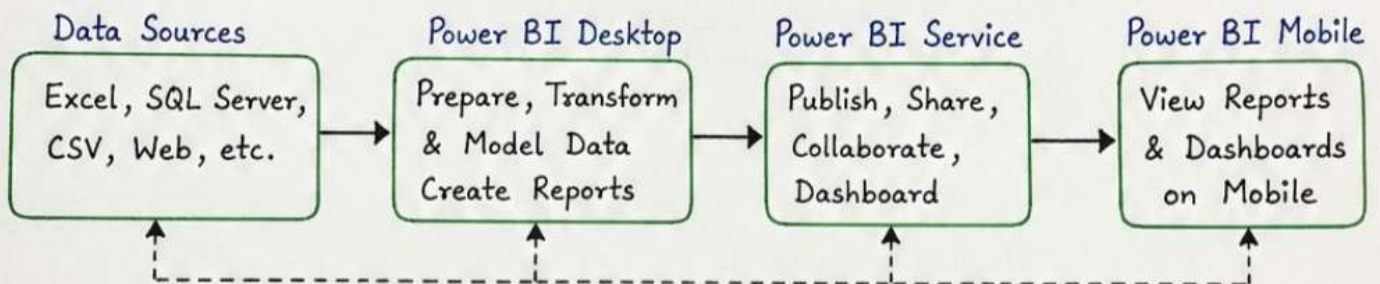
- BI means turning raw data into useful information.

It helps in:

- Data Collection
- Data Analysis
- Data Visualization
- Reporting
- Better Decision Making



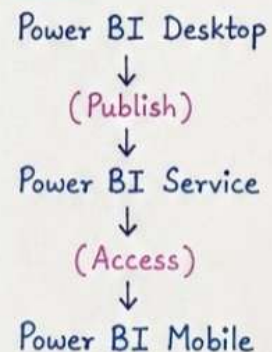
3. Power BI Architecture



4. Power BI Components

- Power BI Desktop :**
It is a free Windows application used to connect, transform, model data and create reports.
- Power BI Service :**
It is an online SaaS (Software as a Service) platform. We can publish reports, create dashboards, share and collaborate here.
- Power BI Mobile :**
It is a mobile application (Android, iOS, Windows) to view reports and dashboards on the go.

Note :



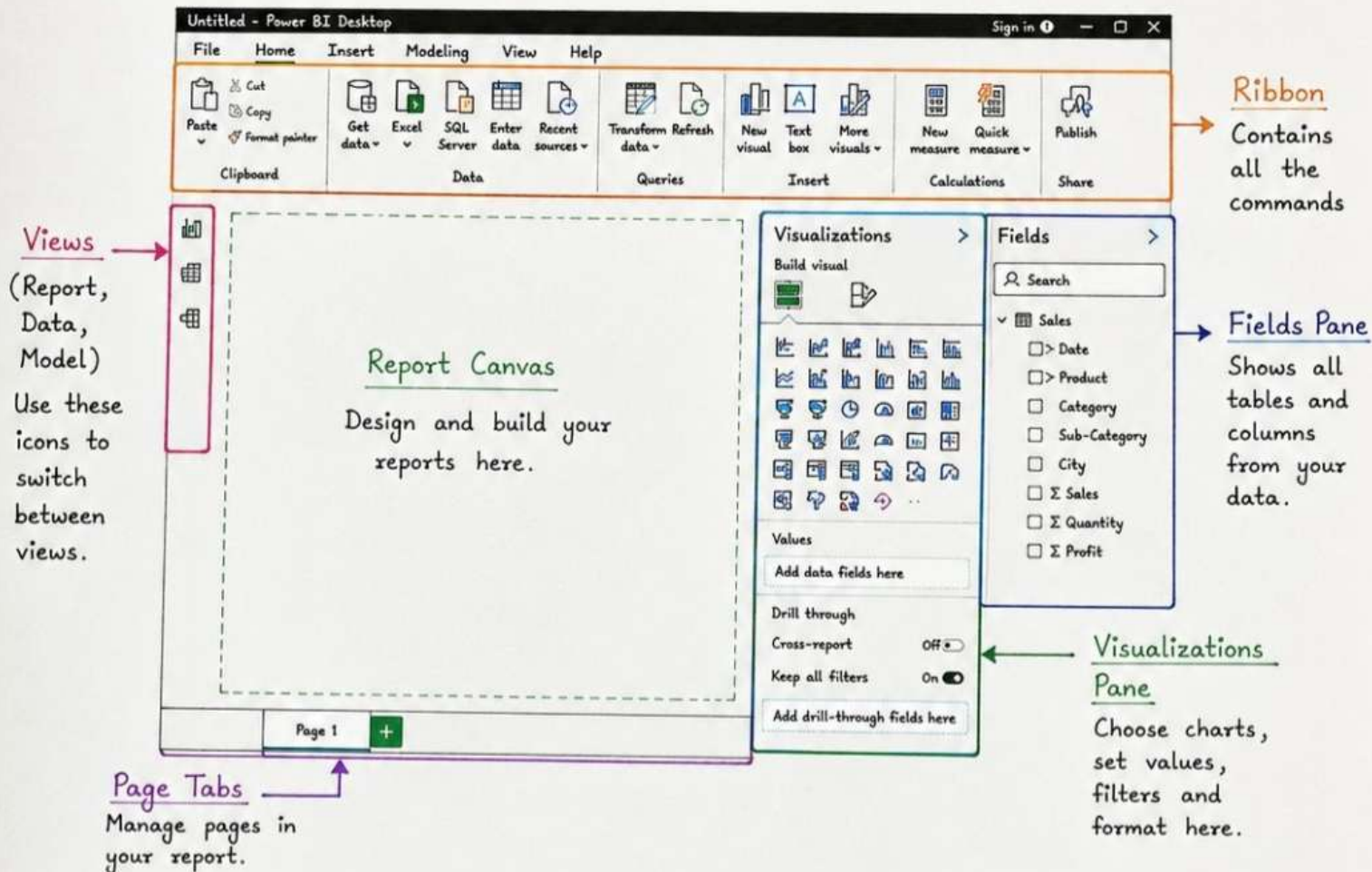
5. Power BI Licensing

- Power BI Free :** Free to use Power BI Desktop and basic sharing.
- Power BI Pro :** For sharing, collaboration and larger datasets.
- Power BI Premium :** For advanced features, large organizations and dedicated capacity.

POWER BI INTERFACE

Power BI Desktop is the main tool used to connect data, transform, model and create reports.

Let's understand the Power BI Desktop interface.



1. Different Views in Power BI Desktop

- Report View** : Used to create interactive reports using visuals.
- Data View** : Used to preview and explore the data in table format.
- Model View** : Used to create relationships between tables and manage the data model.

Tip :

You can switch between views using the icons on the left side.

2. Important Parts

- Ribbon** : Contains all important commands and tools.
- Views** : Helps to switch between Report, Data and Model views.
- Report Canvas** : Area where we design our reports.
- Visualizations Pane** : Used to select chart type and assign fields.
- Fields Pane** : Shows all tables and columns from the data.
- Page Tabs** : Used to add, delete and manage report pages.

Key Takeaway

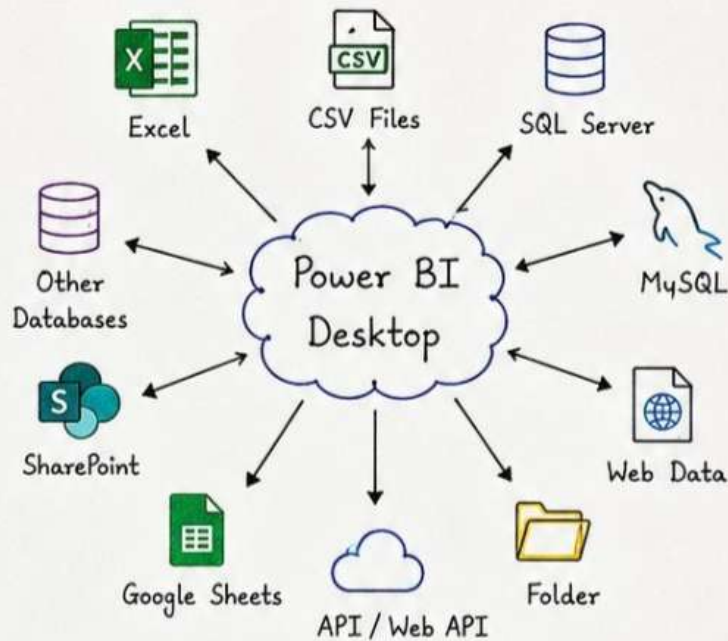
Power BI Desktop interface is simple and user friendly. Understanding each part is the first step to becoming a Power BI expert. ★

DATA SOURCES & DATA IMPORT

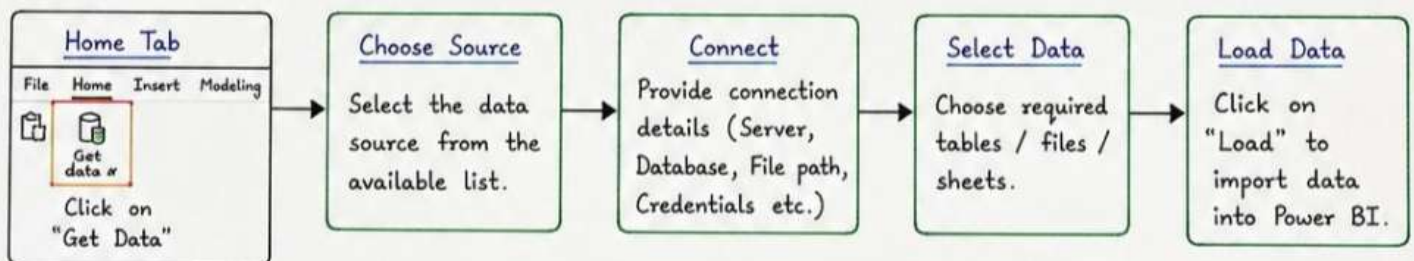
Power BI can connect to a wide variety of data sources. We can import data from these sources and use it to create reports and dashboards.

1. Common Data Sources

- Excel Workbook (.xlsx, .xls)
- CSV Files (.csv)
- Text / Delimited Files
- SQL Server Database
- MySQL Database
- Web Data (Web Page)
- SharePoint Folder / List
- Google Sheets
- Folder (Multiple Files)
- API (Web API)
- Oracle Database
- Azure Services
- Other Databases and Services



2. How to Import Data



3. Data Import Modes

- **Import Mode** : Data is imported and stored in Power BI. Supports fast performance and all Power BI features.
- **DirectQuery Mode** : Data is not imported. Power BI sends queries in real-time to the source. Always up-to-date data but slower.
- **Live Connection** : Connects to Analysis Services / Power BI dataset in real-time. You cannot create calculated columns or tables.

Tip :

For best performance, use Import Mode. Use DirectQuery only for large real-time data.

4. Important Points

- ★ Always clean and prepare data before using in reports.
- ★ Power BI supports scheduled refresh for imported data (in Power BI Service).
- ★ You can combine data from multiple sources.
- ★ Use appropriate data type while importing for better analysis.

Key Takeaway

Power BI can connect to many sources and bring data together in one place for analysis and visualization.



POWER QUERY EDITOR (ETL)

Power Query is a data preparation and transformation tool in Power BI. We use it to clean, shape and transform raw data before loading it into the data model.

1. Power Query Editor Interface

The screenshot shows the Power Query Editor window titled 'Untitled - Power Query Editor'. It features a ribbon with tabs: File, Home, Transform, Add Column, View, and Help. The ribbon contains various icons for data operations. On the left is the 'Queries [2]' pane showing 'Sales' and 'Product' queries. The main area is the 'Data Preview' table. On the right is the 'Query Settings' pane with 'PROPERTIES' and 'APPLIED STEPS' sections.

Annotations:

- Ribbon:** Contains tools and commands.
- Queries Pane:** Shows all queries in your file.
- Data Preview:** Shows the data after each step.
- Query Settings:** Shows properties and all applied steps.

Date	Product	Category	Sales	Quantity
01-01-2024	Pen	Stationery	100	10
02-01-2024	Pencil	Stationery	150	15
03-01-2024	Notebook	Stationery	200	20
04-01-2024	Bag	Accessories	300	5
05-01-2024	Bottle	Accessories	250	10

2. Common Data Transformation Operations

- | | | | |
|--|---|--|--|
| | Remove Columns - Remove unnecessary columns from data. | | Remove Duplicates - Remove duplicate rows. |
| | Remove Rows - Remove unwanted rows (blank, errors, duplicates etc.). | | Append Queries - Add rows from one table to another. |
| | Split Columns - Split a column into multiple columns. | | Merge Queries - Combine two tables based on a common column. |
| | Merge Columns - Combine two or more columns into one. | | Pivot Column - Turn unique values of a column into new columns. |
| | Change Data Type - Change the data type (Text, Number, Date etc.). | | Unpivot Column - Convert columns into attribute-value pairs. |
| | Replace Values - Find and replace specific values. | | Custom Column - Create a new column using custom formula. |

Key Takeaway

Power Query helps us to transform raw data into clean, consistent and analysis ready data. ★

3. Applied Steps

Every transformation you apply is recorded in the Applied Steps list. Steps are executed in order from top to bottom.

- You can edit any step by clicking on the gear icon.
- You can delete, reorder or disable steps.
- Steps can be reused for consistency and automation.

Tip :

Always keep your steps simple, reusable and well organized for better performance and easy maintenance. 😊

DATA MODELING IN POWER BI

Data modeling is the process of creating relationships between tables in Power BI. It helps to organize data and make it easier to analyze.

1. What is Data Model?

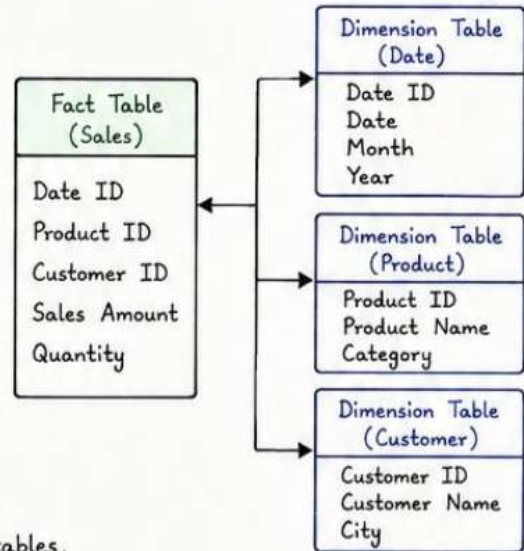
- A data model is a collection of tables, relationships, and calculations (measures and columns).
- It defines how data is organized and how tables are related to each other.

Key Takeaway

Good data model = Accurate reports + Better performance + Easy maintenance. ★

2. Types of Tables

- Fact Table** : Contains measurable data (numbers).
Example: Sales, Orders, Transactions.
- Dimension Table** : Contains descriptive data (details).
Example: Customer, Product, Date, Region.



3. Relationships

- A relationship connects columns from two tables.
- It allows Power BI to combine data from multiple tables.

Types of Relationships:

Type	Diagram	Description	Example
One-to-One (1:1)		One record in table A relates to one record in table B.	Employee table and Employee Detail table (1 employee = 1 detail)
One-to-Many (1:*)		One record in table A relates to many records in table B.	Customer (1) → Sales (Many)
Many-to-Many (*:*)		Many records in table A relate to many records in table B.	Student ↔ Course (Students take many courses and each course has many students)

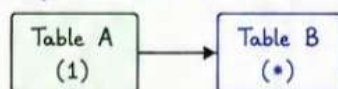
Cardinality

It shows the number of records in one table that relate to records in another table.

1 = One
* = Many

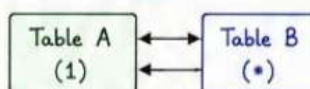
4. Cross Filter Direction

- Single Direction (Default)**



Filter flows from Table A to Table B only.

- Both Direction**



Filter flows in both directions.

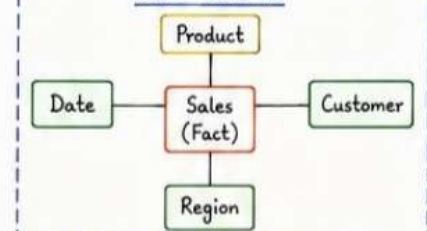
Tip

Use single direction for better performance unless you have a specific requirement. 😊

5. Best Practices for Data Modeling

- ★ Use a Star Schema for better performance and easy reporting.
- ★ Keep fact table in the center and dimension tables around it.
- ★ Use meaningful and consistent names for tables and columns.
- ★ Create relationships using unique key columns.
- ★ Avoid many-to-many relationships if possible.
- ★ Hide unnecessary columns from report view.

Star Schema



DAX (DATA ANALYSIS EXPRESSIONS)

DAX (Data Analysis Expressions) is a formula language used in Power BI to create calculations in tables or models. DAX is used to create Measures, Calculated Columns and Calculated Tables.

1. Types of DAX Calculations

Type	Description	Created In
Measure	A measure is a calculation that is evaluated on the fly based on the filters in the report.	Created in <u>MEASURES</u> (Used in Values area of visuals)
Calculated Column	A calculated column adds new data to a table. It is stored in the model.	Created in <u>COLUMNS</u> (Used like any other column)
Calculated Table	A calculated table creates a new table using a DAX expression.	Created in <u>TABLES</u> (Acts like a normal table in the model)

Key Takeaway

Measures = Dynamic
(changes with filters)
Calculated Column = Static
(stored in model)
Calculated Table = New table
created using DAX



2. DAX Syntax

Basic Syntax

<Name> = <Expression>

Example :

Total Sales = SUM(Sales[Amount])

3. Examples of DAX Calculations

• Measure Example :

Total Sales = SUM(Sales[Amount])

→ This measure calculates the total of Amount column in Sales table.

• Calculated Column Example :

Discount Amount = Sales[Amount] * 0.10

→ This column calculates 10% discount for each row.

• Calculated Table Example :

Top Customers =
TOPN(10, Customers, Customers[Sales], DESC)

→ This table shows top 10 customers by sales.

Operators in DAX

+ Addition
- Subtraction
* Multiplication
/ Division
^ Power
& Concatenation

4. Important DAX Elements

- Table Name : Use TableName[ColumnName]
- Functions : DAX has many built-in functions (SUM, COUNT, IF etc.). Measures use aggregation functions.
- Context : DAX works based on Row Context and Filter Context.

5. Common Aggregate Functions in DAX

Function	Description	Example
SUM()	Adds all values in a column.	Total Sales = SUM(Sales[Amount])
AVERAGE()	Returns the average of values.	Avg Sales = AVERAGE(Sales[Amount])
COUNT()	Counts the number of rows.	Total Orders = COUNT(Sales[Order ID])
DISTINCTCOUNT()	Counts the number of unique values.	Unique Customers = DISTINCTCOUNT(Customers[Customer ID])
MIN()	Returns the minimum value.	Min Sales = MIN(Sales[Amount])
MAX()	Returns the maximum value.	Max Sales = MAX(Sales[Amount])

Note :

DAX is not case sensitive.
Example : sum(), Sum(), SUM() all are valid.
But we should follow consistent naming.



Visualizations help us to represent data in a visual format. Power BI provides a wide range of visuals to analyze and share insights.

1. Commonly Used Visuals in Power BI

Visual	Icon	Description	Best Used For
Table		Displays data in rows and columns.	Detailed data view and quick lookups.
Matrix		Displays data in a hierarchical grid.	Summarized data with subtotals and hierarchies.
Card		Shows a single value.	KPIs, totals, key numbers.
Multi-row Card		Displays multiple values in a card format.	Multiple KPIs in a compact view.
KPI		Shows performance against a goal.	Tracking targets and goals.
Bar Chart		Compares values across categories.	Category comparison (horizontally).
Column Chart		Compares values across categories.	Category comparison (vertically).
Line Chart		Shows trends over a period of time.	Time series data, trends and patterns.
Area Chart		Shows trends with filled area.	Volume over time, cumulative trends.
Pie Chart		Shows parts of a whole.	Percentage distribution (few categories).
Donut Chart		Similar to pie chart with a hole.	Percentage distribution (looks better than pie).
Scatter Chart		Shows relationship between two values.	Correlation, pattern and outliers.
Map		Shows data on a geographical map.	Geographical analysis by country, state, city.
Treemap		Shows hierarchical data in rectangles.	Proportional comparison in hierarchy.
Gauge		Shows value within a range.	Progress towards a specific target.
Waterfall Chart		Shows cumulative effect of values.	Understand how positive and negative values affect the total.
Funnel Chart		Shows stages in a process.	Process flow, conversion analysis.

Tips :

- Choose the right visual to tell the right story.
- Avoid using too many visuals on a single page.
- Keep your visuals simple, clear and focused.
- Use colors and formatting to highlight important insights.



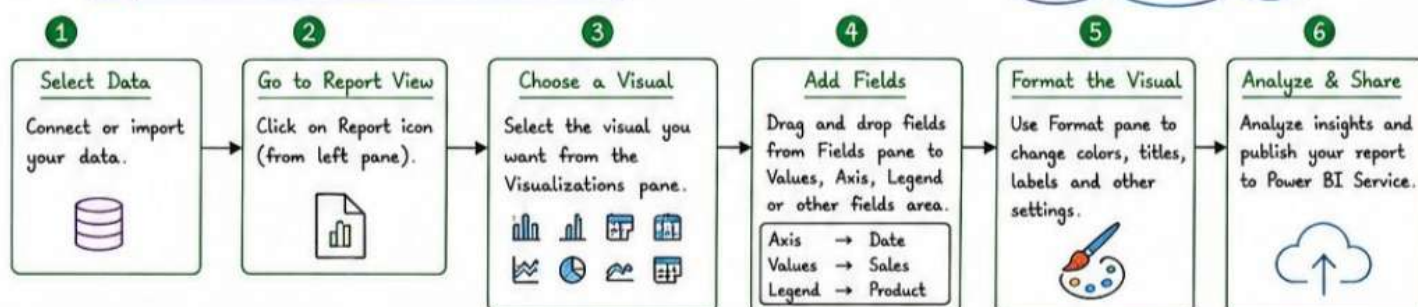
Which Visual to Use?

Compare Categories → Bar/Column Chart
 Show Trend Over Time → Line/Area Chart
 Show Part of Whole → Pie/Donut Chart
 Show Relationship → Scatter Chart
 Show Geographical Data → Map
 Show Hierarchical Data → Treemap/Matrix
 Show Progress to Goal → KPI/Gauge
 Show Flow or Stages → Funnel Chart
 Show Increase/Decrease → Waterfall Chart

Key Takeaway

Visualizations turn your data into meaningful insights. The right visual helps you and your audience understand data quickly and make better decisions. 😊

2. Steps to Add a Visual in Power BI



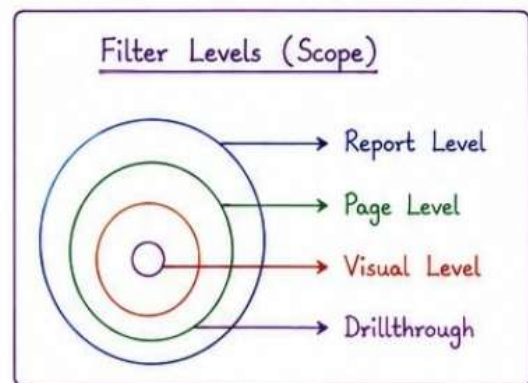
Note : Always design your report keeping the end user in mind. Make it interactive, simple and informative.



Filters and Slicers help us to focus on specific data in a report. They make reports **interactive** and easy to use.

1. Types of Filters in Power BI

1. Visual Level Filter : Filters data for a specific visual only.
2. Page Level Filter : Filters data for the entire page.
3. Report Level Filter : Filters data for all pages in the report.
4. Drillthrough Filter : Passes data to a drillthrough page for more detailed analysis.



Note : Filters reduce the data shown in visuals based on the conditions you apply.

2. Slicers in Power BI

- A slicer is a visual filter.
- Users can select values from slicers to filter data interactively.
- Changes made in a slicer automatically update all connected visuals.

Common Slicer Types

- Dropdown Slicer : Shows values in a dropdown list.
- List Slicer : Shows values in a list with checkboxes.
- Date Slicer : Used to filter dates (Year, Quarter, Month, Day).
- Between Slicer : Used to select a range of numbers or dates.

3. Difference Between Filters and Slicers

Feature	Filters	Slicers
Purpose	Restrict data shown in visuals.	Allow user to interact and filter data.
Visibility	Usually hidden (in Filter pane).	Visible on the report page.
User Interaction	Not directly visible to users.	Users can click and select values.
Types	Visual, Page, Report, Drillthrough.	Dropdown, List, Date, Between.
Best Used For	Page/Report level filtering or advanced conditions.	Quick filtering and interactive analysis.

Tips

- ★ Use slicers for better user experience.
- ★ Sync slicers across pages for consistency.
- ★ Use "Edit Interactions" to control which visuals respond to slicers.

4. Filter Pane in Power BI



- **Filters on this visual** : Applies only to selected visual.
- **Filters on this page** : Applies to all visuals on the current page.
- **Filters on all pages** : Applies to all pages in the report.

5. How to Add a Slicer

- ① Go to Report View.
- ② Click on Slicer icon (in Visualizations pane).
- ③ Drag and drop the field into the slicer field well.
- ④ Choose slicer type from the dropdown (if needed).
- ⑤ Use Format pane to change style, title, etc.
- ⑥ Test the slicer by selecting values.

Slicer Example (Region)

Key Takeaway

Filters and Slicers are powerful tools in Power BI that help users analyze the right data quickly. Use filters for setting rules and slicers for user-friendly interaction.



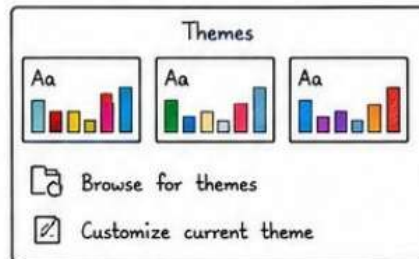
Good formatting and design make your report attractive, easy to read and more effective in delivering insights.

1. Themes in Power BI

- Themes control the overall look and feel of your report.
- You can apply a built-in theme or create a custom theme.
- Theme changes colors, fonts and effects across the report.

Steps to Apply a Theme :

- Go to "View" tab.
- Click on "Themes".
- Choose a theme from the list.
- Your report design will change instantly.



Best Practices

- ★ Use consistent colors throughout the report.
- ★ Keep background simple and clean.
- ★ Use readability-friendly fonts and sizes.
- ★ Highlight important numbers and KPIs. 😊

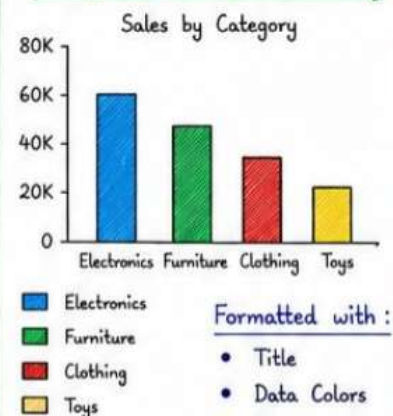
2. Formatting Visuals

You can format each visual to improve clarity and highlight key information.

Common Formatting Options :

Option	Description	Examples / Use
Title	Add or change the title of the visual.	Add meaningful title to charts.
Background	Change background color or add transparency.	Use light colors for better look.
Data Colors	Change colors of data points.	Use different colors for categories.
Data Labels	Show values on charts.	Show value on bars or pie slices.
Axis	Format X-axis and Y-axis.	Change font, color, title, position.
Legend	Show or hide legend and format it.	Change position and text size.
Gridlines	Show or hide gridlines.	Helps in better readability.
Tooltips	Customize tooltip information.	Add more fields in tooltips.
Border & Shadow	Add border or shadow to visual.	Makes visual stand out.

Example (Bar Chart Formatting)



Formatted with :

- Title
- Data Colors
- Data Labels
- Legend
- Gridlines

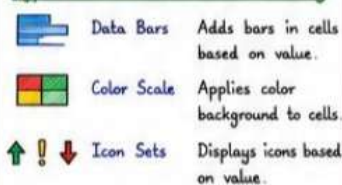
3. Conditional Formatting

- Conditional formatting helps to highlight important data based on conditions.
- It quickly draws attention to key trends and exceptions.

Common Uses :

- Highlight high and low values.
- Show data bars, color scales, icons.
- Make your data easy to understand.

Types of Conditional Formatting



Steps to Apply :

- Select the field or measure.
- Go to "Format" pane.
- Click on conditional formatting (fx).
- Choose the rule and set the conditions.
- Click OK and apply.

Tip

Use conditional formatting for KPIs, targets, performance tracking and exceptions.



4. Additional Design Elements

Titles & Text

- Use clear and meaningful titles.
- Add descriptions or notes if needed.

Sales Dashboard

Overview of Sales Performance

Background

- Use light and simple background.
- Avoid heavy patterns.



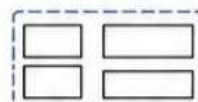
Shapes & Icons

- Use shapes, icons and images to improve understanding.
- Keep it simple.



Alignment

- Align visuals properly.
- Keep equal spacing between elements.



Font & Size

- Use readable fonts.
- Use proper font size for titles, labels and values.

Aa Aa Aa

Key Takeaway

A well formatted and designed report improves readability, grabs attention and helps users understand the insights quickly and take better decisions.



Power BI can connect to many data sources. The "Get Data" feature helps us to import data into Power BI for analysis and reporting.

1. What is a Data Source?

- A data source is the place where data is stored.
- Power BI connects to the data source and brings the data into Power BI.

Key Takeaway

Power BI supports many data sources. Choose the right source based on your data location and business need.



2. Types of Data Sources in Power BI

Category	Examples	Description
File	• Excel Workbook (.xlsx) • CSV, Text Files • XML, JSON Files	Data stored in files on your computer or network.
Database	• SQL Server • MySQL, PostgreSQL • Oracle, IBM DB2	Data stored in relational databases.
Cloud	• Azure SQL Database • Azure Blob Storage • Google BigQuery	Data stored in cloud services.
Online Services	• SharePoint List • Salesforce • Google Analytics	Data available from online platforms.
Other	• Web • OData Feed • Blank Query	Web pages, APIs or other advanced sources.

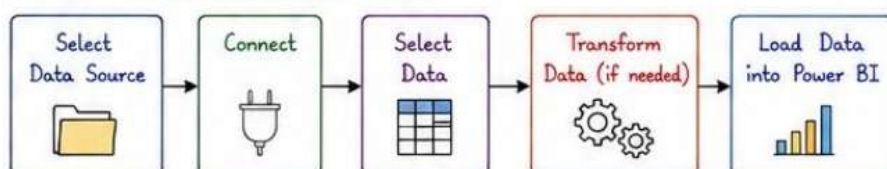
3. Get Data in Power BI

The "Get Data" feature is used to connect and import data into Power BI.

Steps to Get Data :

- ① Open Power BI Desktop.
- ② Go to Home tab.
- ③ Click on Get Data .
- ④ Choose the data source.
- ⑤ Provide connection details if required.
- ⑥ Select the tables or files you want.
- ⑦ Click Load to import data or Transform Data to clean data first.

4. Get Data Flow in Power BI



Tips

- Always preview data before loading.
- Use Transform Data to clean and shape your data.
- Import only the data you need.
- Use appropriate data type for better analysis.



5. Data Import Modes

Mode	Description	Best Used For
Import	Data is imported and stored in Power BI model.	Smaller datasets, fast report performance.
DirectQuery	Data is not stored in Power BI. It queries the source in real time.	Large datasets, real-time data from databases.
Dual	Combination of Import and DirectQuery.	Large models with aggregations.

Import vs DirectQuery

Import	DirectQuery
• Faster performance • Data stored in model • Works offline • Needs data refresh	• Real-time data • Data not stored • Always connected • Slower performance

6. Important Points

- Ensure you have access and permission to the data source.
- Use strong and secure connections.
- Refresh data regularly to keep reports up to date.
- Use appropriate mode (Import/DirectQuery) based on data size and need.
- Clean and transform data for accurate analysis.

Common Data Sources Shortcuts

 Excel : Load Excel files	 SharePoint : Get data from SharePoint lists
 SQL Server : Connect to SQL databases	 Salesforce : Import data from CRM
 Web : Import data from web pages	 Blank Query : Create custom queries

Key Takeaway

Choosing the right data source and using Get Data correctly is the first step towards building powerful and accurate reports in Power BI.



Data modeling organizes data, creates relationships and ensures accurate analysis in Power BI.

1. What is Data Modeling?

- It is the process of organizing data from different sources.
- It involves tables, relationships and calculations.
- It helps in fast queries and accurate results.

2. Components of Data Model

Component	Description	Purpose	Icon
Tables	Collection of related data in rows and columns.	Stores and organizes data.	
Relationships	Connection between tables using matching columns.	Combines data from multiple tables.	
Columns	Fields in a table.	Contains data items.	
Measures	DAX formulas that return a single value.	Performs calculations on data.	
Calculated Columns	New columns created using DAX for each row.	Adds new information to the table.	
Hierarchies	Organizes data in levels.	Enables drill down analysis.	

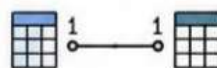
Key Takeaway

A well-designed data model makes reports faster, easier to understand and more reliable.



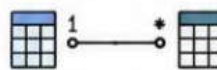
3. Types of Relationships

1. One-to-One (1:1)



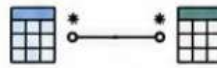
- Each row in Table A relates to one row in Table B.

2. One-to-Many (1:*)



- One row in Table A relates to many rows in Table B.

3. Many-to-Many (*:*)



- Rows in Table A relate to many rows in Table B and vice versa.

4. Creating Relationships

Steps :

- 1 Go to Model view.
- 2 Drag the column from one table to another.
- 3 Power BI automatically detects the relationship.
- 4 Set Cardinality and Cross filter direction.
- 5 Click OK.

Example: One-to-Many Relationship

Customers Table		Sales Table		
CustomerID	Name	SaleID	CustomerID	Amount
1	Amit	101	1	5000
2	Neha	102	1	3500
3	Raj	103	2	2500
		104	3	6000

One customer can have many sales, but each sale belongs to one customer.

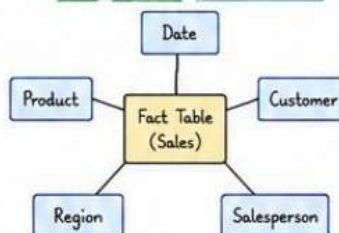
5. Cross Filter Direction

Type	Description	Diagram
Single	Filters flow in one direction (default).	
Both	Filters flow in both directions.	
None	No filter is passed between tables.	

6. Best Practices for Data Modeling

- ✓ Use meaningful table and column names.
- ✓ Remove unnecessary columns and duplicates.
- ✓ Create proper relationships between tables.
- ✓ Use a Date Table for time intelligence.
- ✓ Keep fact tables at the center.
- ✓ Use dimensions to describe facts.
- ✓ Avoid bi-directional relationships unless needed.
- ✓ Test the model for performance.

Star Schema (Recommended)



- Fact table contains measures.
- Dimension tables describe the facts.
- Improves performance and simplifies reports.

7. Role of Date Table

Benefits	Description
Time Intelligence	Enables advanced calculations like YTD, MTD, QTD.
Consistent Analysis	Provides continuous dates even with no data.
Better Filtering	Helps in easy filtering by year, month, quarter, etc.

8. Common DAX in Data Model

DAX Function	Syntax Example	Use
SUM	Total Sales = SUM(Sales[Amount])	Adds values in a column.
COUNTROWS	Total Orders = COUNTROWS(Orders)	Counts number of rows.
CALCULATE	Total PY = CALCULATE([Total Sales], DATEADD('Date'[Date], -1, YEAR))	Changes filter context.
RELATED	City = RELATED(Customer[City])	Gets data from related table.
ALL	All Sales = CALCULATE([Total Sales], ALL(Sales))	Removes filters.

Tip

Good data modeling is the foundation of great reports and dashboards!

Summary

A strong data model connects the right data, in the right way, to deliver the right insights!



DAX is the formula language used in Power BI to create calculations, measures, calculated columns and calculated tables.

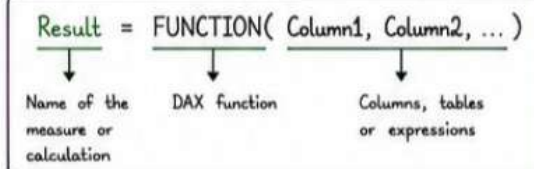
Key Takeaway

- ★ DAX helps you perform powerful calculations and create custom insights in Power BI. 😊

1. Where Can We Use DAX?

Object	Description	Example
Measures	Dynamic calculations that change with filters.	Total Sales = SUM(Sales[Amount]) 
Calculated Columns	New columns added to a table, calculated row by row.	Discounted Price = Sales[Price] * 0.9 
Calculated Tables	New tables created using DAX expressions.	High Sales = FILTER(Sales, Sales[Amount] > 1000) 

2. Basic DAX Syntax



3. Commonly Used DAX Functions

Function	Syntax	Description	Example
SUM	SUM(Column)	Adds all numbers in a column.	SUM(Sales[Amount])
AVERAGE	AVERAGE(Column)	Returns the average of numbers.	AVERAGE(Sales[Amount])
COUNTROWS	COUNTROWS(Table)	Counts the number of rows in a table.	COUNTROWS(Sales)
CALCULATE	CALCULATE(Expression, Filter)	Modifies filter context and evaluates an expression.	CALCULATE(SUM(Sales[Amount]), Sales[Year] = 2024)
FILTER	FILTER(Table, Condition)	Returns a table that satisfies the condition.	FILTER(Sales, Sales[Amount] > 1000)
ALL	ALL(Table/Column)	Removes all filters from table/column.	ALL(Sales)
RELATED	RELATED(Column)	Gets related value from another table.	RELATED(Customer[Country])

Best Practices

- ★ Use measures instead of calculated columns for better performance.
- ★ Use meaningful names for your DAX objects.
- ★ Test your expressions step by step.
- ★ Understand filter context in CALCULATE. ✓

4. Measures Examples

Measure	DAX Expression
Total Sales	Total Sales = SUM(Sales[Amount])
Total Quantity	Total Quantity = SUM(Sales[Quantity])
Average Price	Average Price = AVERAGE(Sales[Price])
Total Orders	Total Orders = COUNTROWS(Sales)
Sales in 2024	Sales in 2024 = CALCULATE(SUM(Sales[Amount]), Sales[Year] = 2024)
Sales Not in 2024	Sales Not in 2024 = CALCULATE(SUM(Sales[Amount]), Sales[Year] <> 2024)
Total Customers	Total Customers = DISTINCTCOUNT(Customer[CustomerID])

5. Calculated Column Example

In Sales Table

Discounted Price = Sales[Price] * 0.9 
 Profit = Sales[Amount] - Sales[Cost]
 Year = YEAR(Sales[OrderDate])

Note

Calculated columns increase the size of your data model. 

6. Calculated Table Example

New Table: High Value Sales

High Value Sales = 
 FILTER(
 Sales,
 Sales[Amount] > 1000
)

Note

Calculated tables are static and created during data refresh. 

7. Filter Context vs Row Context

Filter Context	Row Context
- Determines which rows are considered in a calculation. - Comes from slicers, filters, visuals or CALCULATE. - Example: Total Sales = SUM(Sales[Amount])	- Exists when a formula is evaluated row by row. - Comes from calculated columns or iterators. - Example: Discounted Price = Sales[Price] * 0.9

VS

8. CALCULATE Function in Action

Original Measure

Total Sales =
SUM(Sales[Amount])



With CALCULATE

Sales 2024 =
CALCULATE(
SUM(Sales[Amount]),
Sales[Year] = 2024
)

CALCULATE changes the filter context, so the measure returns result for Year = 2024 only. 

9. Time Intelligence Functions (Examples)

Function	Syntax	Use
TOTALYTD	TOTALYTD(Expression, DateColumn)	Year to date total.
SAMEPERIODLASTYEAR	SAMEPERIODLASTYEAR(DateColumn)	Same period last year.
DATESYTD	DATESYTD(DateColumn)	Returns dates year to date.
DATEADD	DATEADD(DateColumn, -1, YEAR)	Shift dates by a period.

Example

YTD Sales = TOTALYTD([Total Sales], 'Date'[Date])
 Sales Last Year = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date])) 

10. DAX Tips

- ✓ Use variables (VAR) to simplify complex expressions.
- ✓ Use REMOVEFILTERS() or ALL() to clear filters when needed.
- ✓ Use DIVIDE() instead of / to avoid divide by zero errors.
- ✓ Use FORMAT() for formatting numbers or dates.
- ✓ Optimize your DAX for performance. ★

Key Takeaway



DAX is the heart of Power BI analytics. Master DAX to create powerful calculations, deeper insights and advanced reports that drive better business decisions.



Navigation and interactivity make Power BI reports user-friendly, dynamic and easy to explore.

1. Why Navigation & Interactivity?

- Helps users move between pages and visuals.
- Allows filtering and highlighting of data.
- Improves user experience and engagement.
- Makes reports more dynamic and insightful.



2. Types of Navigation

Type	Description	Icon	Example
Page Navigation	Move between pages in a report.		Buttons, Bookmarks
Bookmark Navigation	Save and go to a specific view.		Bookmarks
Drill Navigation	View data in more detail.		Drill down, Drill through
Custom Navigation	Use shapes, images or menus.		Image maps, Menus

Key Takeaway

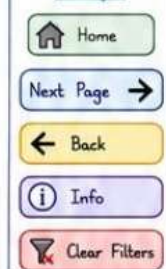
- ★ Good navigation and interactivity help users explore data easily and get better insights. 😊

3. Navigation Using Buttons

Common Button Actions

Action	Description	Icon
Page Navigator	Go to another page.	
Bookmark Navigator	Go to a saved bookmark.	
Back	Go back to previous page.	
Info	Show additional information.	
Clear Filters	Clear all applied filters.	

Example



You can find buttons in Insert tab under Buttons.

4. Bookmarks in Power BI

Steps to Create a Bookmark

- 1 Apply filters, slicers or other changes to your report.
- 2 Go to View tab → Bookmarks pane.
- 3 Click Add to create a bookmark.
- 4 Give it a name.
- 5 Use a button to navigate to the bookmark.

Bookmark Options

- Data (default)
- Display
- Current Page
- All Visuals



Bookmarks help users quickly switch between different views of the report.

5. Drill Down, Drill Up & Drill Through

Drill Down	Drill Up	Drill Through
View data at a lower level of detail.	Go back to a higher level of detail.	Navigate to a different page with more details.
Category → Subcategory	Subcategory → Category	From summary to detailed page

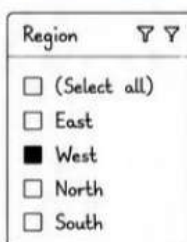
How to Enable Drill Through

- 1 Create a new detailed page.
- 2 Add fields that act as Drill Through fields.
- 3 Go to Model view and select the field.
- 4 In Properties pane → Turn on Drill Through.
- 5 Now, users can right-click a data point and select Drill through.

Properties

General	▼
Name	CustomerID
Drill through	On

6. Slicers for Interactivity



Slicer Tips

- ✓ Use slicers to filter visuals on the page.
- ✓ Supports single select and multi-select.
- ✓ Use dropdown, list or between styles.
- ✓ Sync slicers across pages for consistency.
- ✓ Add slicer titles for better understanding.



Slicers make reports interactive and user-driven.

7. Sync Slicers Across Pages

Steps

- 1 Add a slicer to a page.
- 2 Go to View tab.
- 3 Click Sync slicers.
- 4 Check the pages where you want to sync.
- 5 Click OK.

Sync Slicers Panel

Page Name	Visible	Sync
Overview		☒
Sales		☒
Products		☐
Region Analysis		☒

Visible: Show slicer on page

Sync: Apply slicer selection to page

8. Report Tooltips

Tooltips display additional information when users hover over a visual.

How to Create a Tooltip Page

- 1 Create a new page.
- 2 Turn off Page Navigator.
- 3 Design the tooltip content.
- 4 Go to Page Information → Tooltip (On).
- 5 Hover over the visual to see the tooltip.

Example



Tooltip Best Practices

- ✓ Keep it clean and simple.
- ✓ Show only key information.
- ✓ Use meaningful titles and proper formatting.
- ✓ Improve user experience.

9. Other Interactivity Features

Highlight



Highlight data points across visuals.

Cross Filter



Filter related visuals by selecting data.

Q&A Visual



Ask questions in natural language and get answers.

Smart Narratives



Automatically explains insights in your data.

Personalize Visuals



Allow users to personalize their visuals.

Key Takeaway

Good navigation and interactivity turn reports into powerful storytelling tools, helping users explore data, discover insights and take action with confidence.



Dashboards give a quick overview of key insights. Publishing makes your reports accessible and shareable in the Power BI Service.

Key Takeaway

- ★ Dashboards provide an at-a-glance view of important data and publishing helps you share insights securely with others. 😊

1. Dashboard in Power BI

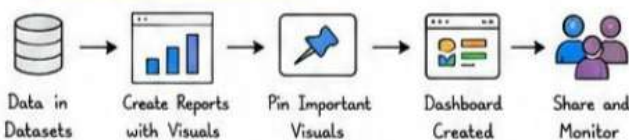
- A dashboard is a single page view of important visuals from one or more reports.
- It is ideal for monitoring KPIs and business metrics.
- Dashboards are interactive and update with real-time data.



2. Dashboard vs Report

Feature	Dashboard	Report
Purpose	Monitor KPIs and key metrics.	Analyze and explore detailed data.
Data Source	Can use multiple reports.	Based on a single dataset.
Interactivity	High-level interactivity.	Detailed drill and exploration.
Layout	Single page.	Multiple pages.
Best For	Quick insights and monitoring.	In-depth analysis and storytelling.

3. How Dashboards Work



💡 You can pin visuals from multiple reports to a single dashboard.

Tip

Keep dashboards simple, focused and updated. Too many visuals can make it cluttered.

4. Publishing a Report

Steps to Publish:

- Click File → Publish.
- Select a destination Workspace.
- Click Select and then Publish.
- Wait for the upload to complete.
- View the report in Power BI Service.



5. Workspaces in Power BI Service

- Workspaces are containers to store, manage and share reports, dashboards and datasets.
- They help in collaboration and access management.

Types of Workspaces	
Type	Description
My Workspace	Personal workspace for individual use.
Team Workspace	Shared workspace for team collaboration.
App Workspace	Created when an app is published.

6. Sharing Options

Method	Description	Use Case
Share Dashboard	Share a dashboard with users or groups.	Quick sharing of insights.
Share Report	Share a report with specific permissions.	Detailed access control.
Publish to Web	Embed and share with anyone (public).	For public or external audience.
Mobile Access	Access dashboards and reports on mobile devices.	On-the-go insights.

7. Refresh Data in Service

- Keep your dashboards and reports up-to-date by refreshing data.
- You can schedule refresh or refresh manually.

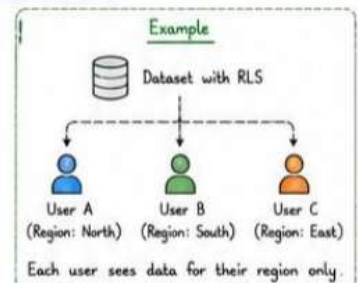
Refresh Methods	
Scheduled Refresh	Set a schedule (e.g., daily, hourly) to refresh data automatically.
Manual Refresh	Refresh data manually whenever needed.

Note

Scheduled refresh requires a data gateway if your data source is on-premises.

8. Row-Level Security (RLS) in Service

- RLS allows you to restrict data access based on user roles.
- Users see only the data they are allowed to see.



9. Dashboard Tiles

- Dashboard is made up of tiles.
- Each tile represents a visual or KPI.
- You can customize size, position and filtering.

- Tile Actions**
- ✓ Resize
 - ✓ Move
 - ✓ Filter
 - ✓ Drill-through
 - ✓ Delete

Tile Types	
Visual Tile (Chart, Table, etc.)	
KPI Tile	
Text Tile	
Image Tile	
Web Content Tile	

10. Best Practices

- ✓ Design dashboards with a clear purpose.
- ✓ Show only key metrics and insights.
- ✓ Use consistent colors and formatting.
- ✓ Keep dashboards uncluttered and easy to read.
- ✓ Set up scheduled refresh for latest data.
- ✓ Manage access and security properly.
- ✓ Test dashboards on different devices.



A great dashboard tells the story at a glance, and publishing shares that story with the right audience!



Power Query Editor is the heart of data transformation in Power BI.

It helps you clean, shape and prepare data before loading it to the model.

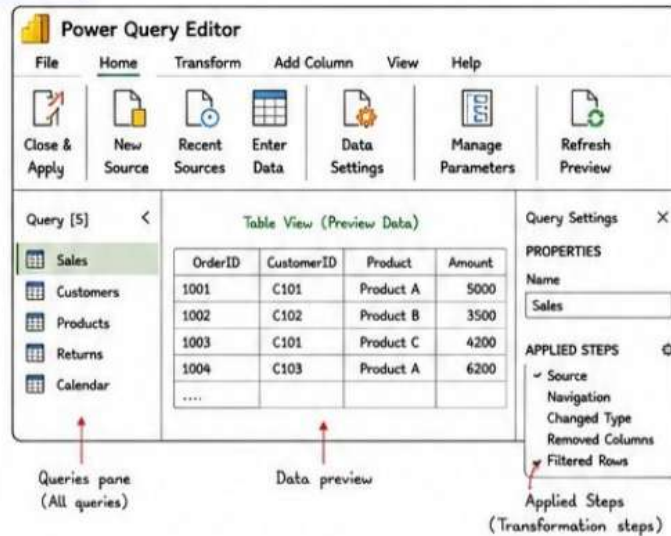
Key Takeaway

- ☆ Power Query helps convert raw data into clean, reliable and analysis-ready data.



1. What is Power Query Editor?

- It is a data connection and transformation tool.
- You can clean, combine and reshape data without writing any code.
- All steps are recorded in the Applied Steps pane.

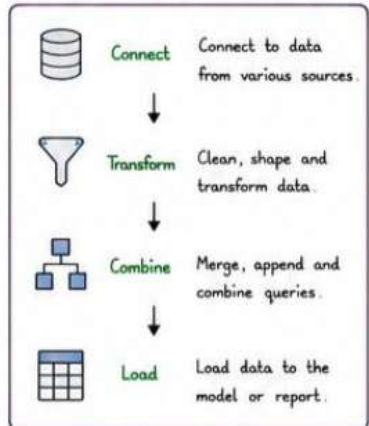


Queries pane
(All queries)

Data preview

Applied Steps
(Transformation steps)

2. Power Query Workflow



3. Common Data Sources

Category	Examples	Icon
Files	Excel, CSV, Text/CSV, JSON, XML, Folder	
Databases	SQL Server, Access, Oracle, MySQL, PostgreSQL	
Online Services	SharePoint, OneDrive, Web, OData, Salesforce	
Other	Blank Query, Power BI Datasets, R, Python	

4. Common Transformations

Transformation	Description
Remove Rows	Remove unwanted rows.
Remove Columns	Remove unnecessary columns.
Change Data Type	Change the data type of a column.
Split Column	Split one column into multiple columns.
Merge Columns	Combine multiple columns into one.
Filter Rows	Keep rows that meet a condition.
Sort Rows	Sort data in ascending or descending order.
Replace Values	Replace old values with new values.
Fill	Fill down, up or fill blank values.
Pivot/Unpivot	Reshape data for better analysis.

5. Applied Steps

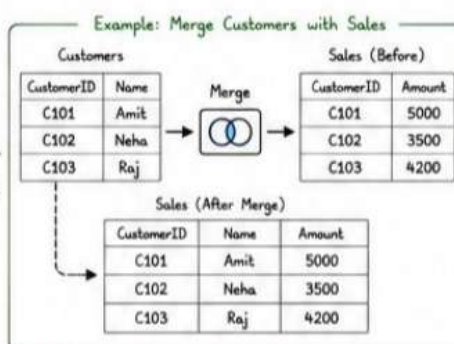
- Every action in Power Query is recorded as a step.
- Steps are applied from top to bottom.
- You can edit, delete, reorder or insert steps.

Tips

- ✓ Rename queries meaningfully.
- ✓ Remove unnecessary steps.
- ✓ Keep transformations efficient.
- ✓ Preview data at each step.
- ✓ Use data types correctly.

6. Merge Queries

- Merge combines columns from one query to another based on a related column.
- It works like a VLOOKUP.



7. Append Queries

- Append stacks rows from two or more queries with the same columns.
- It works like UNION ALL.



8. Useful Menu Tabs

Tab	Key Features	Icon
Home	New Source, Manage Queries, Close & Apply	
Transform	Data type, Remove/Keep Rows & Columns, Split, Group, Replace Values	
Add Column	Custom Column, Conditional Column, Index Column, Merge Column	
View	Query Settings, Advanced Editor, Data Preview	

9. Best Practices

- ✓ Load only clean and necessary data.
- ✓ Remove unused columns and rows early.
- ✓ Apply filters as early as possible.
- ✓ Use appropriate data types.
- ✓ Use reference queries to reuse steps.
- ✓ Document your steps and logic.
- ✓ Keep queries simple and maintainable.

10. Keyboard Shortcuts

Action	Shortcut
Close & Apply	Alt + F4
Advanced Editor	Alt + F11
Go to Step	Ctrl + G
Delete Step	Delete
Move Step Up	Ctrl + Up Arrow
Move Step Down	Ctrl + Down Arrow
Show/Hide Preview	Alt + P

Summary



Power Query Editor transforms raw data into valuable insights by cleaning, shaping, combining and preparing data the right way!



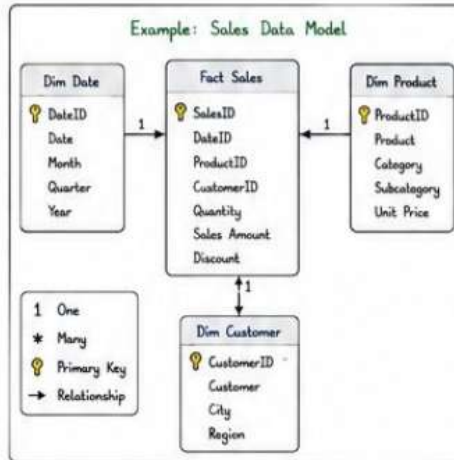
Data modeling organizes data into tables and relationships to create a powerful foundation for reports and analysis.

Key Takeaway

- ★ A well-designed data model improves performance, ensures data accuracy and makes reports easy to build and maintain. 😊

1. What is Data Modeling?

- Data modeling defines how data objects (tables) relate to each other.
- It helps represent real-world business logic.
- A good model is simple, accurate and efficient.



2. Types of Relationships

Relationship	Description	Example	Cardinality
One-to-Many	One record in a table relates to many records in another table.	Customer to Sales	1 → ∞
Many-to-One	Many records relate to one record.	Sales to Customer	* → 1
One-to-One	A record in a table relates to one record in another table.	User to User Details	1 ↔ 1
Many-to-Many	Many records in a table relate to many records in another table.	Product to Promotion	* ↔ *

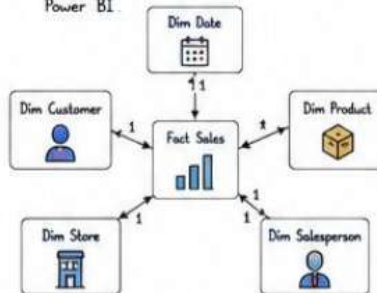
3. Cardinality and Cross Filter Direction

Cardinality	Cross Filter Direction
1 One: Unique values in the 'one' side.	Single (Default)
* Many: Multiple values in the 'many' side.	Both

💡 Use Single direction unless you have a specific reason to use Both (can impact performance).

4. Star Schema (Recommended)

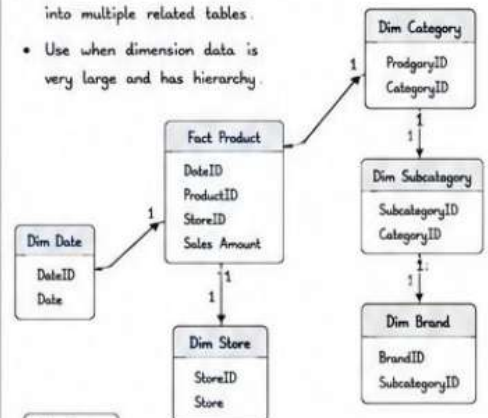
- A star schema has a central fact table connected to dimension tables.
- It is easy to understand, performant and ideal for Power BI.



✅ Prefer more dimensions and fewer facts. Avoid snowflakes unless necessary.

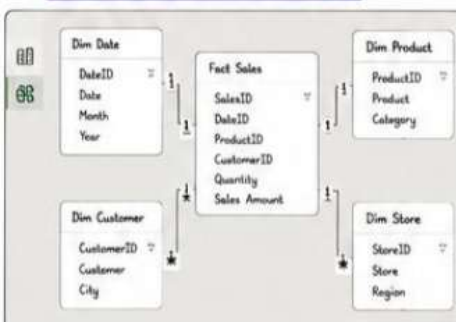
5. Snowflake Schema (When Needed)

- A dimension table is normalized into multiple related tables.
- Use when dimension data is very large and has hierarchy.



1 One
* Many

6. Data Model View in Power BI



7. Best Practices for Data Modeling

- ✓ Use a star schema.
- ✓ Use meaningful and consistent names.
- ✓ Keep fact tables at the center.
- ✓ Use surrogate keys (integers) for relationships.
- ✓ Hide unused columns.
- ✓ Avoid bi-directional relationships unless required.
- ✓ Keep data types appropriate.
- ✓ Remove duplicate and unnecessary data.
- ✓ Document your model.

8. Tips

- ★ Always relate dimensions to facts, not to each other.
- ★ Use date dimension to enable time intelligence.
- ★ Create calculation groups when needed for advanced scenarios.
- ★ Check model performance using Performance Analyzer.

9. Common Issues and Fixes

Issue	Cause	Fix
Incorrect totals	Many-to-many or wrong relationships	Review relationships and cardinality
Slow performance	Bi-directional Filters or large tables	Use single direction and optimize model
Duplicate rows	Missing or incorrect relationships	Check keys and relationships
Missing data	Inactive relationships	Activate relationship or use DAX (USERELATIONSHIP)

10. Helpful DAX Functions for Modeling

Function	Use
RELATED()	Get related column value from another table.
RELATEDTABLE()	Return related rows from another table.
LOOKUPVALUE()	Find a value in a table based on criteria.
USERELATIONSHIP()	Use an inactive relationship in a calculation.

11. Checklist

- ✓ All tables have primary keys.
- ✓ Relationships are correct.
- ✓ Model follows star schema.
- ✓ Data types are correct.
- ✓ No unnecessary columns.
- ✓ Model is tested with reports.



A strong data model is the foundation of great Power BI reports. Model it right, and your insights will be powerful and reliable!



DAX is the formula language of Power BI. It helps you create calculations, measures, and columns to perform advanced analysis.

Key Takeaway

- ★ DAX turns your data into intelligent insights with powerful calculations. 😊

1. What is DAX?

- DAX (Data Analysis Expressions) is a formula language.
- It is used to create measures, calculated columns and tables.
- DAX formulas always start with an = sign.
- DAX is case-insensitive.

Example DAX Formula

= SUM(Sales[Amount])

Starts with = Function Table[Column]

2. Types of DAX Calculations

Type	Description	Example
Measure	Calculated at query time. Changes with filters.	Total Sales = SUM(Sales[Amount])
Calculated Column	Calculated at data refresh time. Stored in the model.	Profit = Sales[Amount] - Sales[Cost]
Calculated Table	Creates a new table using DAX.	Top Customers = FILTER(...)

Tips



- ★ Use measures for dynamic results.
- ★ Use columns for row level calculations.
- ★ Use tables for advanced scenarios.

3. Basic DAX Syntax

= [Function] [Argument1, Argument2, ...]

Structure

- Functions perform operations.
- Arguments can be columns, values or expressions.

4. Common DAX Functions

Category	Function	Description	Example
Aggregation	SUM()	Adds all values.	= SUM(Sales[Amount])
	COUNT()	Counts rows.	= COUNT(Sales[OrderID])
	AVERAGE()	Average of values.	= AVERAGE(Sales[Amount])
Filtering	CALCULATE()	Changes filter context.	= CALCULATE([Total Sales], Year = 2024)
	FILTER()	Filters a table.	= FILTER(Sales, Sales[Amount] > 1000)
Logical	IF()	Returns value based on condition.	= IF(Sales[Amount] > 1000, "High", "Low")
Time Intelligence	TOTALYTD()	Year-to-date total.	= TOTALYTD([Total Sales], Date[Date])
	SAMEPERIODLASTYEAR()	Compares with last year.	= SAMEPERIODLASTYEAR(Date[Date])

5. Creating a Measure

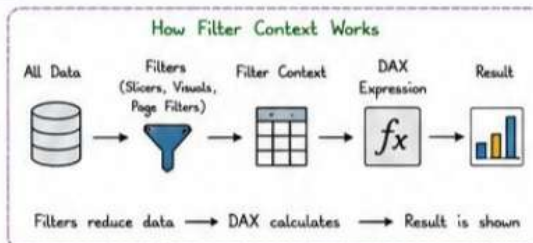
- Go to Modeling tab.
- Click New Measure.
- Write DAX formula.
- Press Enter.
- Use the measure in visuals.

Example:

Total Sales =
SUM(Sales[Amount])

6. Filter Context in DAX

- Filter context defines the data visible to a DAX expression.
- It can come from slicers, visuals, or page/report filters.
- CALCULATE() can modify filter context.



7. CALCULATE() Example

Scenario: Total Sales for 2024 only

Original Measure

Total Sales = SUM(Sales[Amount])

With CALCULATE()

Total Sales 2024 =
CALCULATE([Total Sales],
Date[Year] = 2024)

Modifies filter context
Filter condition

8. Row Context vs Filter Context

Row Context	Filter Context
Exists in calculated columns and iterators.	Exists in measures and CALCULATE().
Operates on one row at a time.	Works on a set of rows.
Created automatically.	Created by filters or DAX functions.
Used for row-level calculations.	Used for result-level calculations.

9. Time Intelligence Example

Goal: Sales Year-to-Date

Sales YTD =
TOTALYTD([Total Sales], Date[Date])

Goal: Sales Same Period Last Year

Sales LY =
CALCULATE([Total Sales],
SAMEPERIODLASTYEAR(Date[Date]))

- ★ Time intelligence functions work with a proper Date table. 📅

10. DAX Best Practices

- ✓ Use meaningful names.
- ✓ Prefer measures over calculated columns.
- ✓ Keep DAX formulas simple and readable.
- ✓ Use variables (VAR) for complex formulas.
- ✓ Avoid unnecessary CALCULATE().
- ✓ Test and validate your results. 🎯

11. Useful DAX Operators

Operator	Description	Example
+	Add	= Sales[Amount] + Sales[Tax]
-	Subtract	= Sales[Amount] - Sales[Cost]
*	Multiply	= Sales[Quantity] * Sales[Price]
/	Divide	= Sales[Profit] / Sales[Sales]
&	Concatenate	= Sales[FirstName] & " " & Sales[LastName]
=	Equal	= Sales[Region] = "North"
>	Greater Than	= Sales[Amount] > 1000
<	Less Than	= Sales[Amount] < 1000

12. Sample DAX Measures

Measure Name	DAX Formula
Total Sales	= SUM(Sales[Amount])
Total Profit	= SUM(Sales[Profit])
Average Order Value	= DIVIDE([Total Sales], COUNT(Sales[OrderID]))
Orders Count	= DISTINCTCOUNT(Sales[OrderID])
Profit Margin %	= DIVIDE([Total Profit], [Total Sales])
Sales vs LY	= [Total Sales] - CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Date[Date]))

13. Remember

- ★ DAX is powerful but learn step by step.
- ★ Understand filter context deeply.
- ★ Practice with real data.
- ★ Great reports start with great DAX! 👍



Master DAX, master insights!

The more you practice, the more powerful your reports become.



Measures and Columns are both important in Power BI.

Knowing when to use each one helps you build **efficient** and **correct** models.

Key Takeaway

- Use Columns for storing data.
- Use Measures for calculations.
- Use the right one for better performance.



1. What are Columns?

- Columns store data in each row of a table.
- Calculated Columns are added during data refresh.
- They increase the size of the data model.
- Used for filtering, grouping, sorting and relationships.

Example: Column in a Table

OrderID	Product	Quantity	Unit Price	Total (Column)
1001	Laptop	2	500	1000
1002	Mobile	1	300	300
1003	Tablet	1	200	200
1004	Laptop	1	500	500
...	...	...	...	...

Calculated Column Example:
Total = 'Sales'[Quantity] * 'Sales'[Unit Price]

2. What are Measures?

- Measures are calculated on the fly.
- They do not increase the size of the model.
- Used in values, KPIs and calculations.
- Results change with filters and slicers.

Example: Measure Result

Product	Total Sales (Measure)
Laptop	1,500
Mobile	300
Tablet	200
Grand Total	2,000

Measure Example:
Total Sales = SUM('Sales'[Quantity] * 'Sales'[Unit Price])

3. Key Differences

Aspect	Column	Measure
Calculation Time	Calculated at data refresh time	Calculated at query time
Storage	Stored in the data model	Not stored, calculated on the fly
Size Impact	Increases model size	Does not increase model size
Used In	Rows, Filters, Slicers, Groups	Values, KPIs, Cards, Charts
Changes with Filters	No (static per row)	Yes (dynamic)
Best For	Storing data and row-level logic	Aggregations and business calculations

4. When to Use?

Use Columns When:

- You need the value in each row.
- You need to filter or group by it.
- You need it in relationships.
- You use it in slicers.
- The logic is row-based and static.

Use Measures When:

- You need dynamic calculations.
- The result depends on filters.
- You are aggregating data.
- You want to save model size.
- You use it in charts and KPIs.

5. Examples Side by Side

Sales Table

Date	Product	Quantity	Unit Price
01-Jan	Laptop	2	500
01-Jan	Mobile	1	300
01-Jan	Tablet	1	200
02-Jan	Laptop	1	500

Calculated Column (Stored)

Total (Column) =
'Sales'[Quantity] *
'Sales'[Unit Price]

Result in Table

1000
300
200
500

Measure (Dynamic)

Total Sales (Measure) =
SUM('Sales'[Quantity] *
'Sales'[Unit Price])

Result in Visual (by Date)

Date	Total Sales
01-Jan	1500
02-Jan	500
Grand Total	2000

6. Common Scenarios

Scenario	Use	Example
Discount % for each product	Column	Discount % = 'Products'[Discount] / 100
Total Sales by Year	Measure	Total Sales = SUM('Sales'[Quantity] * 'Sales'[Unit Price])
Age from Date of Birth	Column	Age = DATEDIFF('Customers'[DOB], TODAY(), YEAR)
Running Total	Measure	Running Total = CALCULATE([Total Sales], FILTER(ALL('Date'[Date]), 'Date'[Date] <= MAX('Date'[Date])))
Profit Margin %	Measure	Profit Margin % = DIVIDE([Total Profit], [Total Sales])
Category Grouping	Column	Category Group = IF('Products'[Category] = "Tech", "Technology", "Other")

7. Performance Impact

Columns (More Impact on Model Size)

- Stored in memory.
- More columns = larger model.
- Slower refresh if too many.

Measures (Less Impact)

- Calculated on the fly.
- Do not increase model size.
- Depends on data size and complexity.



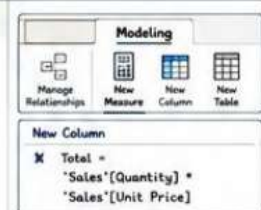
Best Practice

Store data in Columns,
do calculations in Measures!

8. How to Create

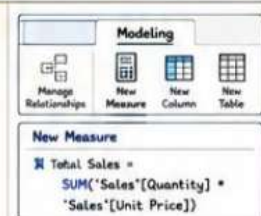
Create a Column

- Select the table.
- Click Modeling tab.
- Click New Column.
- Write DAX formula.
- Press Enter.



Create a Measure

- Select the table.
- Click Modeling tab.
- Click New Measure.
- Write DAX formula.
- Press Enter.



9. Do's and Don'ts

Do's

- Use columns for row-level data.
- Use measures for aggregations.
- Use meaningful names.
- Keep DAX simple and readable.
- Optimize for performance.

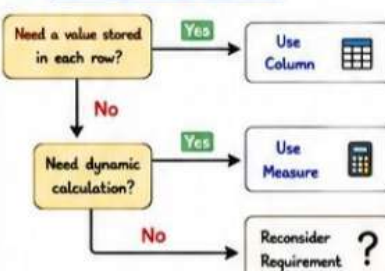


Don'ts

- Don't create unnecessary columns.
- Don't use columns for totals.
- Don't use complex DAX in columns.
- Don't forget filter context in measures.
- Don't store data you can calculate.



10. Quick Decision Guide



11. Summary

- Columns store data.
- Measures calculate results.
- Use the right one for the right need.
- Better models give better reports!



Right Choice + Good DAX = Powerful Insights!



Key Takeaway

- ★ Dashboards provide a single view of your key insights and help you monitor what matters most 😊

Dashboards in Power BI let you monitor important metrics, track KPIs and make data-driven decisions – anytime, anywhere.

1. What is a Dashboard?

- A dashboard is a single page view of important visuals.
- It shows KPIs and key metrics at a glance.
- Dashboards are ideal for monitoring and tracking performance.



2. Dashboard vs Report

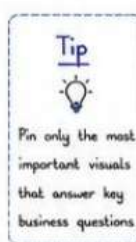
Feature	Dashboard	Report
Purpose	Monitor & track KPIs	Explore & analyze data
Layout	Single page (tile-based)	Multi-page (detailed)
Interactivity	Limited (Filters)	High (Filters, slicers, drill)
Data Refresh	Real-time (tiles)	On demand / schedule
Best For	Overview & monitoring	Deep dive & analysis

3. Dashboard Components

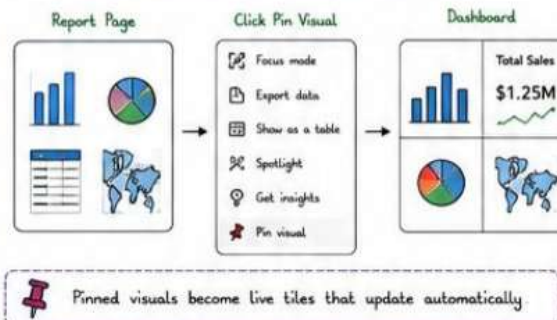
Component	Description
Tiles	Visuals, KPIs, metrics
Charts	Show trends & comparisons
KPIs	Track goals & performance
Filters	Narrow down the data
Text / Images	Add notes & context
Buttons	Navigate or take action

4. Creating a Dashboard

- Go to a report.
- Pin important visuals to a dashboard.
- Arrange and resize tiles.
- Add text, images or buttons.
- Apply filters (optional).
- Save and share the dashboard.



5. Pinning Visuals to a Dashboard



6. KPIs and Goals

KPI Example

Total Sales
\$1.25M
Goal: \$1.50M
83%

KPIs help you track progress toward goals.

KPI Indicators

- On Track ($\geq 100\%$)
- Near Target (70% - 99%)
- Off Track ($< 70\%$)

7. Dashboard Filters

Filter types you can use:

Time Range	Filter by relative or specific dates.
Filters Pane	Add page-level or tile-level filters.
Slicers (from report)	Pin slicers for interactive filtering.
Cross-Filter	Click one tile to filter others (if enabled).

8. Sharing and Collaboration

Share Dashboard
Share with users or groups.

Publish to Web
Make dashboard public.

Subscribe
Get data alerts via email.

Export
Export tiles or dashboard.

Best Practice

- ✓ Share with the right audience.
- ✓ Use row-level security (RLS).
- ✓ Provide clear context and instructions.

9. Mobile Dashboards

Access your dashboards on the go!



Tip

Design dashboards that are mobile-friendly and focus on key metrics.

10. Real-Time Dashboards

- Use real-time data sources (Streaming datasets, IoT, Azure services, etc.).
- Monitor live KPIs and operational metrics.
- Great for scenarios like operations, support, web analytics, and more.



11. Dashboard Best Practices

- ✓ Keep it simple and focused.
- ✓ Show only what matters.
- ✓ Use meaningful titles and labels.
- ✓ Use consistent colors and formatting.
- ✓ Limit the number of visuals.
- ✓ Use KPIs for key metrics.
- ✓ Review and update regularly.

12. Common Use Cases

Use Case	Example
Sales	Sales performance, pipeline, revenue by region
Marketing	Campaign performance, leads, website analytics
Finance	Revenue, expenses, profit, budget vs actual
Operations	Inventory, orders, delivery, support tickets
HR	Headcount, attrition, training performance

13. Sample Dashboard Layout



14. Checklist

- ✓ Do I have the right KPIs?
- ✓ Are the visuals clear and easy to understand?
- ✓ Is the dashboard simple and focused?
- ✓ Have I added meaningful titles and labels?
- ✓ Are filters and interactions working?
- ✓ Is it mobile-friendly?
- ✓ Have I tested and shared the dashboard?



A great dashboard turns data into action – keep it simple, keep it focused, and keep improving!

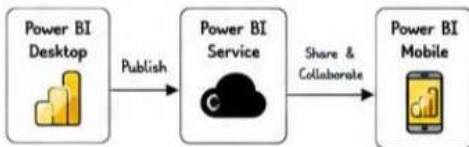


Key Takeaway

- ★ Power BI Service lets you publish, share, collaborate and manage content in the cloud. 😊

Power BI Service is the online platform to publish your reports, share insights and collaborate securely.

1. Power BI Ecosystem



- Create reports in Power BI Desktop.
- Publish to Power BI Service (cloud).
- Share and collaborate with others.
- Access on the go with Power BI Mobile.

2. Publish to Power BI Service

- ① Click "Publish" in Power BI Desktop.
- ② Sign in to your Power BI account.
- ③ Choose a Workspace.
- ④ Your report is now available in the Service.

★ Tip

Organize content in Workspaces for better management. 💡

3. Workspaces

Type	Use
My Workspace	Personal workspace for individual use.
Team Workspace	Collaborate with a team or group.
Premium Workspace	For datasets >1GB and advanced features.
App Workspace (Premium)	Distribute content at scale with apps.

4. Sharing Options

Option	Description	Best For
Share with individuals	Share reports or dashboards with specific people.	One-to-one collaboration
Share with groups	Share with Microsoft 365 Groups.	Team collaboration
Publish to app	Distribute content to a wider audience.	Organization-wide sharing
Share link	Generate a link to view the report or dashboard.	Quick sharing (Use with care!)

5. Permissions in Workspaces

Role	Permissions
Admin	Full control (Add members, delete content, manage settings)
Member	Create, edit, and share content within the workspace.
Contributor	Add content and edit existing content.
Viewer	View content only



Best Practice

Give the least privilege needed for the task.

6. Apps in Power BI

- ✓ Bundle dashboards, reports and datasets into an App.
- ✓ Distribute the App to users or groups.
- ✓ Keep content up-to-date centrally.



7. Data Refresh in the Service

- Datasets need refresh to show latest data.
- Configure scheduled refresh or refresh on demand.
- Gateway is required for on-premises data sources.

Refresh Options	
Scheduled Refresh	Set times for automatic refresh.
Refresh Now	Manually refresh the dataset.
Gateway Refresh	Use On-premises Data Gateway for local data.

8. Alerts and Subscriptions

Stay informed!



Alerts

Get notified when data meets conditions.

Subscriptions

Schedule emails of reports or dashboards as snapshots.

Go to the report or dashboard → Set up alerts or subscriptions.

9. Security in Power BI Service



Row-Level Security (RLS)
Restrict data access based on user.



Sensitivity Labels
Classify and protect your content.



Azure AD Integration
Secure sign-in and user management.



Audit & Activity Logs
Track user activities and changes.

10. Mobile Access



- ✓ Access dashboards and reports on the go.
- ✓ Optimized experience on Power BI Mobile app.
- ✓ Stay connected to key insights anytime.

★ Tip

Pin important dashboards to your mobile home.

11. Collaboration Best Practices

- ✓ Use Workspaces to organize content.
- ✓ Use clear names and descriptions.
- ✓ Share selectively and securely.
- ✓ Keep data and reports up-to-date.
- ✓ Communicate insights with your team.



12. Common Tasks in Service

Task	How to Do
Pin a visual	Open report → Pin (pushpin icon) → Select dashboard.
Favorite content	Click ★ on a report or dashboard.
Clone report	Click → Save a copy.
Export data	From a visual → Export data.
Embed report	File → Embed report → Copy link or code.

13. Checklist Before Sharing

- ✓ Is the data accurate and up-to-date?
- ✓ Are visuals clear and easy to understand?
- ✓ Are filters and slicers working properly?
- ✓ Are permissions set correctly?
- ✓ Have I tested the report end-to-end?



14. Remember

- ★ Publish to the cloud.
- ★ Share the right way.
- ★ Collaborate securely.
- ★ Keep content fresh.
- ★ Drive decisions together!



15. Key Takeaway



Power BI Service brings your data to the cloud, connects your team to insights, and helps your organization make better decisions—together!



Publish with confidence. Share with purpose. Collaborate for impact! ★